

MATH 3220-002 PSet 10

Specification

Instructor: Alp Uzman

Subject to Change; Last Updated: 2025-11-14 22:10:22-07:00

1 Background

This problem set focuses on differential forms. There are three perspectives one can consider differential forms from, namely:

Synthetic: Differential forms are certain strings of symbols subject to certain operational rules.

Differential Calculus: Differential forms are alternating multilinear function valued functions.

Integral Calculus: Differential forms are objects whose signed integrals over chains are well defined.

In this problem set, all problems develop the synthetic approach.

Synthetic Perspective on Differential Forms Here is a short summary of the synthetic approach to differential forms. Let $M \subseteq \mathbb{R}^n$ be an open subset. Then a differential form on M is a finite linear combination of real valued function multiples of finitely many of

$$1, dx_1, dx_2, \dots, dx_m$$

wedged together. Addition and scalar multiplication work as expected. The wedge operation is compatible with these vector space operations, but it also satisfies additional rules, described below.

Collections of Differential Forms The collection of all differential forms on M is denoted by $\text{Form}(M)$. If only C^k real valued functions are allowed in each summand, the resulting subcollection is denoted by $C^k\text{-Form}(M)$. If only exactly p distinct $dx_1, dx_2, \dots, dx_m$'s are allowed in each summand, the resulting subcollection is denoted by $\text{Form}^p(M)$. Finally we put $C^k\text{-Form}^p(M) = C^k\text{-Form}(M) \cap \text{Form}^p(M)$. All these collections are vector spaces.

Thus for instance

- Any C^k differential 0-form is of the form

$$\omega = f,$$

where $f \in C^k(M; \mathbb{R})$.

- Any C^k differential 1-form is of the form

$$\omega = f_1 dx_1 + f_2 dx_2 + \dots + f_n dx_n,$$

where $f_1, f_2, \dots, f_n \in C^k(M; \mathbb{R})$.

- Any C^k differential 2-form is of the form

$$\omega = f_{12} dx_1 \wedge dx_2 + f_{13} dx_1 \wedge dx_3 + \dots + f_{n-1n} dx_{n-1} \wedge dx_n,$$

where $f_{12}, f_{13}, \dots, f_{n-1n} \in C^k(M; \mathbb{R})$.

- Any C^k differential 3-form is of the form

$$\begin{aligned}\omega &= f_{123} dx_1 \wedge dx_2 \wedge dx_3 \\ &\quad + f_{124} dx_1 \wedge dx_2 \wedge dx_4 \\ &\quad + \cdots \\ &\quad + f_{n-2n-1n} dx_{n-2} \wedge dx_{n-1} \wedge dx_n,\end{aligned}$$

where all coefficient functions are in $C^k(M; \mathbb{R})$.

- Any C^k differential form is of the form

$$\omega = \omega_0 + \omega_1 + \omega_2 + \cdots + \omega_m,$$

where m is a nonnegative integer, and for each $p \in \{0, 1, \dots, m\}$, ω_p is a C^k differential p -form. As ω_i 's are additive components of ω , instead of using the "product" symbol $\prod$, one uses the "direct sum" symbol $\oplus$ and writes

$$\begin{aligned}\text{Form}(M) &= \bigoplus_{p=0}^{\infty} \text{Form}^p(M), \\ C^k\text{-Form}(M) &= \bigoplus_{p=0}^{\infty} C^k\text{-Form}^p(M)\end{aligned}$$

to summarize the state of affairs.

Wedge Product The wedge product on differential forms is a bilinear transformation

$$\wedge : \text{Form}(M) \times \text{Form}(M) \rightarrow \text{Form}(M)$$

subject to the following additional rules:

WedgeRule1: For any two $\omega, \eta \in \text{Form}(M)$ and for any $f \in \text{Form}^0(M)$,

$$f(\omega \wedge \eta) = (f\omega) \wedge \eta = \omega \wedge (f\eta) = f \wedge \omega \wedge \eta.$$

WedgeRule2: For any $i \in \{1, 2, \dots, n\}$,

$$dx_i \wedge dx_i = 0.$$

WedgeRule3: For any $i, j \in \{1, 2, \dots, n\}$,

$$dx_i \wedge dx_j = -dx_j \wedge dx_i.$$

In particular, if $\omega \in \text{Form}^p(M)$ and $\eta \in \text{Form}^q(M)$, then $\omega \wedge \eta \in \text{Form}^{p+q}(M)$. As wedge products do not affect differentiability, if $\omega \in C^k\text{-Form}^p(M)$ and $\eta \in C^\ell\text{-Form}^q(M)$, then $\omega \wedge \eta \in C^{\min\{k, \ell\}}\text{-Form}^{p+q}(M)$. Note that by the first rule above, while one can put a wedge between a function (0-form) and a differential form, as the position of the function does not matter, often this wedge is suppressed.

Exterior Derivative The exterior derivative on differential forms is a linear transformation

$$d : C^{k+1}\text{-Form}(M) \rightarrow C^k\text{-Form}(M)$$

subject to the following additional rules:

ExtRule1: For any $f \in C^{k+1}\text{-Form}^0(M)$,

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \cdots + \frac{\partial f}{\partial x_n} dx_n = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

ExtRule2: For any $f \in C^{k+1}\text{-Form}^0(M)$ and any $i_1, \dots, i_p \in \{1, \dots, n\}$,

$$d(f dx_{i_1} \wedge dx_{i_2} \wedge \cdots \wedge dx_{i_p}) = (df) \wedge dx_{i_1} \wedge dx_{i_2} \wedge \cdots \wedge dx_{i_p}.$$

In particular if $\omega \in C^{k+1}\text{-Form}^p(M)$, then $d\omega \in C^k\text{-Form}^{p+1}(M)$.

Exact versus Closed Differential forms The exterior differentiation operator d is a linear transformation, hence its kernel $\ker(d)$ is a vector subspace of $C^1\text{-Form}(M)$ and its image $\text{im}(d)$ is a vector subspace of $\text{Form}(M)$.

- A differential form ω is **exact** if it is in the image of the exterior differentiation operator d , that is, if

$$\exists \phi \in C^1\text{-Form}(M) : d\phi = \omega.$$

- A differential form ω is **closed** if it is in the kernel of the exterior differentiation operator d , that is,

$$d\omega = 0.$$

- Two differential forms ω and η are C^k -**cohomologous** if their difference is a C^k exact differential form, that is,

$$\exists \phi \in C^{k+1}\text{-Form}(M) : \omega = \eta + d\phi.$$

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details. The [hyperlinks](#) too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to).

However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook; it's up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. There is nothing to prove in this problem. In this problem we consider differential forms on $\mathbb{R}^3$.

- (a) Compute $d\omega$, where $\omega = x^2dx + xydy$.
- (b) Compute $\omega \wedge \eta$, where $\omega = x^2dx + xydy$, and $\eta = ydx + x^3dy$.
- (c) Compute $\omega \wedge \eta \wedge \zeta$, where $\omega = zdx + xdy + ydz$, $\eta = \cos zdx + \sin zdy + xydz$ and $\zeta = d(xyz)$.

2. **There is nothing to prove in this problem.** In this problem we consider differential 1-forms on $\mathbb{R}^2$. For any $v = (v_1, \dots, v_6) \in \mathbb{R}^6$, define

$$\omega_v = (v_1 + v_2x + v_3y)dx + (v_4 + v_5x + v_6y)dy.$$

- (a) Compute the set of all $v \in \mathbb{R}^6$ such that ω_v is exact.
 (b) Compute the set of all $v \in \mathbb{R}^6$ such that ω_v is closed.
3. In this problem we consider differential 1-forms on $\mathbb{R}^2$.

- (a) Let $\phi \in C^0(\mathbb{R}; \mathbb{R})$ and define

$$\omega_\phi = \phi(x)ydx + dy.$$

Then for any function $f \in C^1(\mathbb{R}; \mathbb{R})$, $f\omega$ is exact iff

$$\exists x_0 \in \mathbb{R}, \forall x \in \mathbb{R} : f(x) = \exp \left(\int_{x_0}^x \phi(u)du \right).$$

- (b) For any two functions $\phi, \psi \in C^0(\mathbb{R}; \mathbb{R})$, the following differential 1-form on $\mathbb{R}^2$ is exact:

$$\omega = \phi(x)dx + \psi(y)dy.$$

- (c) For any $\omega = Pdx + Qdy \in C^1\text{-Form}^1(\mathbb{R}^2)$, if ω is exact, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

4. Let $M \subseteq \mathbb{R}^n$ be open.

- (a) For any $i \in \{1, 2, \dots, n\}$, dx_i is a closed form.
- (b) For any $i \in \{1, 2, \dots, n\}$, dx_i is an exact form.
- (c) For any $f \in C^2\text{-Form}^0(M)$ and any $i_1, i_2, \dots, i_p \in \{1, 2, \dots, n\}$,

$$dd(fdx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_p}) = 0.$$

- (d) For any $\omega \in C^2\text{-Form}(M)$, $dd\omega = 0$.
- (e) For any $\omega \in C^2\text{-Form}(M)$, if ω is exact, then it is also closed.
- (f) For any $\omega \in C^2\text{-Form}(M)$, if ω is closed, then it is also exact.

5. Let $M \subseteq \mathbb{R}^n$ be open and $\omega \in \text{Form}^k(M)$.

- (a) If k is odd, then $\omega \wedge \omega = 0$.
- (b) If $2k \geq n$, then $\omega \wedge \omega = 0$.
- (c) If $2k \leq n$ and $\omega \wedge \omega = 0$, then k is odd.

6. Let $M \subseteq \mathbb{R}^n$ be open, $\omega \in C^1\text{-Form}^p(M)$, and $\eta \in C^1\text{-Form}^q(M)$.

- (a) $\omega \wedge \eta = (-1)^{pq}\eta \wedge \omega$.
- (b) $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p\omega \wedge (d\eta)$.

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a **generative AI tool** or a **computer algebra system** for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or
- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., [ChatGPT](#)) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the [Wayback Machine](#) (see their ["Save Page Now" documentation](#)) to create a permanent, timestamped archive of your chat. There are also third-party browser extensions that are designed to extract chat logs from generative tools.
- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as **prompt engineering**, and it is your responsibility you design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

Here are some guides regarding prompt engineering:

- <https://platform.openai.com/docs/guides/prompt-engineering>
- <https://developers.google.com/machine-learning/resources/prompt-eng>
- <https://www.ibm.com/topics/prompt-engineering>
- <https://aws.amazon.com/what-is/prompt-engineering/>

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/QrxV4DMrJfdzrhNa8>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

[https://www.gradescope.com/courses/1073533/assignments/6451793,](https://www.gradescope.com/courses/1073533/assignments/6451793)

see the Gradescope [documentation](#) for instructions.

You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

This problem set is due on November 21, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.